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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

MICRO PROJECT REPORT

VibeGuard: An Embedded Vibration Condition Monitoring System

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of

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of

A P J ABDUL KALAM TECHNOLOGICAL UNIVERSITY

CREATING TECHNOLOGY LEADERS OF TOMORROW

OCTOBER 2026

DECLARATION

We hereby declare that the report on "**VibeGuard: An Embedded Vibration Condition Monitoring System**", submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the supervision of **Ms. Sagna L T**. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in this submission. We understand that any violation of the above will be a cause for disciplinary action by the Institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously used by anybody as a basis for the award of any degree, diploma or similar title of any other University.

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BONAFIDE CERTIFICATE

This is to certify that the seminar report entitled **VibeGuard: An Embedded Vibration Condition Monitoring System** submitted by **AMITH KRISHNA DAS, ARCHA PRAMOD, NIHAD P C, SREEHARI K, SREEPRADA K S** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of Bachelor of Technology degree in Computer Science & Engineering(Cyber Security) is a bonafide record of the Micro Project carried out by them under our supervision and guidance.

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- **PEO 3 :** Graduates will demonstrate ethical responsibility, leadership, teamwork, and effective communication while adapting to evolving global technological environments through lifelong learning.

PROGRAMME SPECIFIC OBJECTIVES (PSOs)

- **PSO 1 :** Apply cyber security principles to mitigate cyber risk and analyze security requirements of an organization.
- **PSO 2 :** Develop and deploy secure software and infrastructure using modern tools and industry standards.
- **PSO 3 :** Uphold ethical, legal, and societal responsibilities while protecting privacy, data, and digital assets.

Course Outcomes (COs)

Course: PBCST504 – Microcontrollers

- **CO1 :** Explain the architectural features and instructions of the ARM microcontrollers. (K2)
- **CO2 :** Develop applications involving interfacing of external devices and I/O with ARM microcontroller. (K3)
- **CO3 :** Use various communication protocols of interaction with peer devices and peripherals. (K3)
- **CO4 :** Demonstrate the use of a real time operating system in embedded system applications. (K3)
- **CO5 :** Apply hardware security features of ARM in real world applications. (K3)

PO - CO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
PBCST504.1	3	2	1	1	3	-	-	-	-	-	-	2	2	-
PBCST504.2	3	2	3	2	3	1	-	-	1	1	1	2	3	1
PBCST504.3	3	2	2	2	3	1	-	-	2	1	1	2	3	1
PBCST504.4	3	2	3	2	3	1	-	-	2	1	1	2	3	1
PBCST504.5	3	3	3	2	3	2	1	2	1	1	2	3	3	2

Correlation Levels: 3 – Substantial (High), 2 – Moderate (Medium), 1 – Slight (Low), '-' – No correlation

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Amith Krishna Das

Archa Pramod

Nihad P C

Sreehari K

Sreeprada K S

SUSTAINABLE DEVELOPMENT GOALS (SDGs)

1. **No Poverty** – End poverty in all its forms everywhere.
2. **Zero Hunger** – End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.
3. **Good Health and Well-being** – Ensure healthy lives and promote well-being for all at all ages.
4. **Quality Education** – Ensure inclusive and equitable quality education and promote life-long learning opportunities for all.
5. **Gender Equality** – Achieve gender equality and empower all women and girls.
6. **Clean Water and Sanitation** – Ensure availability and sustainable management of water and sanitation for all.
7. **Affordable and Clean Energy** – Ensure access to affordable, reliable, sustainable and modern energy for all.
8. **Decent Work and Economic Growth** – Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.
9. **Industry, Innovation, and Infrastructure** – Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.
10. **Reduced Inequalities** – Reduce inequality within and among countries.
11. **Sustainable Cities and Communities** – Make cities and human settlements inclusive, safe, resilient, and sustainable.
12. **Responsible Consumption and Production** – Ensure sustainable consumption and production patterns.

13. **Climate Action** – Take urgent action to combat climate change and its impacts.

14. **Life Below Water** – Conserve and sustainably use the oceans, seas and marine resources for sustainable development.

15. **Life on Land** – Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, and halt biodiversity loss.

16. **Peace, Justice, and Strong Institutions** – Promote peaceful and inclusive societies for sustainable development, provide access to justice for all.

17. **Partnerships for the Goals** – Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development.

SDG ADDRESSED BY THE MICROPROJECT AND JUSTIFICATION

PROJECT TITLE	**VibeGuard: An Embedded Vibration Condition Monitoring System**
SDG 9	**SDG 9: Industry, Innovation, and Infrastructure** • Develops a low-cost, edge-resident vibration condition monitoring solution for rotating electromechanical machinery. • Integrates a 32-bit ESP32 microcontroller and ADXL345 3-axis accelerometer for local real-time anomaly detection without costly cloud dependencies. • Supports predictive maintenance and continuous machine health supervision, bolstering resilient manufacturing infrastructure and preventing catastrophic mechanical breakdowns.
SDG 12	**SDG 12: Responsible Consumption and Production** • Extends industrial motor, bearing, and pump lifespans through early identification of mechanical unbalance and excessive vibration wear. • Drastically minimizes industrial e-waste and premature machine scrapping by enabling proactive, targeted servicing rather than whole-assembly replacement. • Promotes ultra-low-power local computing, operating under 1.5 W continuous power to foster energy-efficient industrial sensing.
SDG 4	**SDG 4: Quality Education** • Delivers hands-on, practical engineering laboratory exposure in microcontroller interfacing, real-time operating systems, SPI protocol communication, and digital signal processing. • Bridges theoretical mechanical vibration concepts with embedded edge firmware implementation. • Cultivates ethical, reproducible research methodologies and rigorous hardware verification among undergraduate engineering students.

ABSTRACT

Vibration-based condition monitoring serves as a vital cornerstone for ensuring the operational safety, mechanical integrity, and predictive maintenance of rotating electromechanical machinery. This project presents the design, embedded implementation, and empirical validation of **VibeGuard**, a low-cost, edge-resident vibration condition monitoring system. VibeGuard acquires triaxial acceleration data at an 800 Hz Output Data Rate (ODR) from an Analog Devices ADXL345 MEMS sensor interfaced with a 32-bit dual-core 7Semi ESP32-DEVKIT-E microcontroller over a deterministic 5 MHz 4-wire SPI bus. The microcontroller executes an autonomous digital signal processing pipeline directly at the edge, performing continuous in-place zero-mean DC gravity cancellation, 3-axis Vector RMS (V_{RMS}) feature extraction, and temporal K-of-M persistence filtering ($K = 4$ violations out of $M = 5$ consecutive 320 ms windows) without relying on external cloud dependencies.

To validate the system, an experimental testbed comprising a 12 V DC micro metal gear motor running at 600 RPM (10.0 Hz rotational shaft frequency) and an eccentric 3D-printed rotor arm was evaluated across multiple controlled operational stages. Under healthy baseline conditions (Stage 0), the system established an autonomous reference with $V_{\mathrm{RMS}} \approx 0.013$–$0.060\,g$, illuminating a solid Green LED. When a moderate unbalance mass was introduced (Stage 1: bolt only, $m = 1.2$ g), the composite vibration rose to 0.82–0.86 g, decisively asserting a solid Yellow warning state with zero spurious alarm transitions. Under severe mechanical unbalance (Stage 2/3: bolt plus locknuts, $m = 1.85$ g), dynamic vibration surged to 1.079 g (a 423% increase), triggering a high-visibility 6.25 Hz Fast Red Strobe alert within 1.28 seconds. On-demand 256-point Hann-windowed Fast Fourier Transform (FFT) analysis confirmed a 16-fold spectral energy concentration centered precisely at the 10.0 Hz synchronous 1X mechanical unbalance peak. Screwdriver impulse shock testing confirmed complete transient rejection

($V = 1 < 4$, zero false alarms). Telemetry was streamed to a local web-based dashboard for real-time visualization. Consuming under 1.5 W, VibeGuard demonstrates an explainable, robust, and cost-effective paradigm for industrial asset protection and embedded engineering education.

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LIST OF ABBREVIATIONS

ADC	Analog-to-Digital Converter
ADXL345	Analog Devices 3-Axis Digital MEMS Accelerometer
ARM	Advanced RISC Machines
BOM	Bill of Materials
CS	Chip Select (SPI bus signal)
DC	Direct Current
DSP	Digital Signal Processing
EMF	Electromotive Force (Back-EMF)
ESP32	Espressif 32-bit Dual-Core Microcontroller
FFT	Fast Fourier Transform
GPIO	General-Purpose Input/Output
IIoT	Industrial Internet of Things
JSON	JavaScript Object Notation
KTU	A P J Abdul Kalam Technological University
LDO	Low Drop-Out Voltage Regulator
LED	Light Emitting Diode
MAD	Median Absolute Deviation
MCU	Microcontroller Unit
MEMS	Micro-Electro-Mechanical Systems
MISO	Master In Slave Out (SPI bus signal)
MOSI	Master Out Slave In (SPI bus signal)
ODR	Output Data Rate
PCB	Printed Circuit Board
PEO	Programme Educational Objective
PO	Programme Outcome
PSO	Programme Specific Outcome
RGB	Red-Green-Blue (Tri-color Indicator LED)
RMS	Root Mean Square
RPM	Revolutions Per Minute
SCLK	Serial Clock (SPI bus signal)

LIST OF NOTATIONS

$a_x[n], a_y[n], a_z[n]$	Raw acceleration samples along X, Y, and Z axes (g)
$\tilde{a}_x[n], \tilde{a}_y[n], \tilde{a}_z[n]$	Zero-mean DC-removed acceleration samples (g)
μ_x, μ_y, μ_z	Gravitational DC static offset estimations per axis (g)
V_{RMS}	Triaxial composite Vector Root Mean Square acceleration (g)
N	Number of samples per processing window ($N = 256$)
f_s	Accelerometer Output Data Rate sampling frequency ($f_s = 800\,\text{Hz}$)
T_w	Temporal duration of an analysis window ($T_w = 320\,\text{ms}$)
f_{rot}	Mechanical motor shaft rotational frequency ($f_{\text{rot}} = 10.0\,\text{Hz}$ at 600 RPM)
f_k	Frequency represented by discrete FFT bin index k (Hz)
Δf	Frequency bin resolution ($\Delta f = f_s/N = 3.125\,\text{Hz}$)
$m_{\text{unbalance}}$	Dynamic eccentric unbalance calibration mass (g)
r	Pitch radius of eccentric mass mounting hole ($r = 12.0\,\text{mm}$)
F_c	Dynamic centrifugal force generated by unbalance mass (N)
K, M	Temporal persistence filter parameters ($K = 4$ violations out of $M = 5$ windows)
$J(t)$	Composite anomaly severity index
T_{warn}	Warning condition threshold level (g)
T_{alarm}	Critical alarm condition threshold level (g)

CHAPTER 1
INTRODUCTION

1.1 Overview

Rotating machines produce vibration as a natural consequence of motion, balance, mounting, load, and dynamic structural interaction. When an electromechanical operating condition deteriorates due to mechanical imbalance, misalignment, or fastener looseness, the characteristic vibration pattern changes. VibeGuard is an autonomous, edge-resident vibration condition monitoring system engineered to transform triaxial accelerometer signals from a rotating mechanical testbed into an explainable, real-time health assessment directly on a low-cost embedded microcontroller.

The system interfaces an Analog Devices ADXL345 3-axis digital MEMS accelerometer with a 32-bit dual-core 7Semi ESP32-DEVKIT-E microcontroller over a high-speed 5 MHz SPI bus operating at an 800 Hz Output Data Rate (ODR). The onboard firmware executes continuous in-place zero-mean DC gravity cancellation, statistical Vector Root Mean Square (V_{RMS}) feature extraction, and a deterministic K-of-M temporal persistence filter. A triggered 256-point Hann-windowed Fast Fourier Transform (FFT) provides harmonic spectral verification, while a common-cathode RGB status LED and a host-side serial/WebSocket web dashboard provide immediate local visual alerting and real-time telemetry streaming [1].

1.2 Motivation

The motivation behind the VibeGuard project originates from critical practical challenges identified in industrial asset maintenance and embedded systems education:

1. **Cost and Accessibility Barriers:** Commercial industrial vibration-monitoring instru-

ments often cost hundreds to thousands of dollars per channel, placing automated condition monitoring out of reach for small workshops, light industrial machinery, and undergraduate engineering laboratories.

2. **Need for Quantitative, Measurable Awareness:** Vibrations represent the earliest physical symptom of rotating mechanical degradation. Having continuous, objective mathematical measurements allows maintenance teams to intercept minor issues before they escalate into structural failure.

3. **Limitations of Subjective Manual Inspection:** Traditional tactile or acoustic checks (listening with a mechanic's screwdriver or stethoscope) are subjective, dangerous around high-speed machinery, and fail to provide timestamped quantitative records for predictive trending.

4. **The Edge Computing Opportunity:** Modern low-cost microcontrollers such as the ESP32 possess sufficient dual-core processing power to execute real-time digital filtering, windowed statistics, and spectral transforms locally, enabling self-contained, cloud-free condition monitors.

1.3 Objective

The primary objective of VibeGuard is to design, implement, and empirically validate an autonomous embedded vibration monitoring testbed capable of establishing a healthy operating baseline and reliably distinguishing between normal operation, moderate unbalance warnings, and severe unbalance alarms. The core technical objectives are:

1. To acquire deterministic triaxial acceleration data at 800 Hz from the ADXL345 accelerometer via a dedicated 4-wire SPI bus.

2. To execute in-place zero-mean DC offset removal to eliminate static gravitational vectors

from dynamic vibration analysis.

3. To extract 3-axis composite Vector RMS (V_{RMS}) and implement an adaptive autotuning threshold ladder (Median + $k \times$ MAD).
4. To apply a temporal persistence filter ($K = 4$ violations out of $M = 5$ consecutive 320 ms windows) to eliminate false alarms caused by transient mechanical shocks.
5. To integrate an on-demand 256-point Hann-windowed FFT pipeline for verifying synchronous 1X mechanical unbalance harmonics (10.0 Hz).
6. To provide clear, multi-state visual alerts via an RGB LED (Solid Green, Solid Yellow, Fast Red Strobe, Solid Blue, Purple) and live telemetry to an interactive web dashboard.
7. To evaluate the system under controlled laboratory conditions across three physical stages (Bare Arm, Bolt Only, and Bolt + Nut unbalance).

1.4 Problem Statement

In industrial manufacturing, rotating equipment such as induction motors, blowers, and pumps are vulnerable to sudden dynamic unbalance, leading to catastrophic equipment failure, unscheduled production stoppages, and severe economic losses. Conventional industrial condition monitoring systems typically rely on expensive piezoelectric accelerometers ($>$ \$500), proprietary high-end data acquisition hardware, or continuous high-bandwidth cloud streaming architectures that suffer from network latency and high recurring subscription costs.

Conversely, simple edge thresholding approaches often suffer from frequent false alarms caused by static gravity tilt or non-critical transient mechanical jolts. VibeGuard addresses this engineering challenge by delivering an entirely edge-resident, explainable, and cost-effective monitoring platform that achieves sub-1.3 second detection latency, eliminates external cloud dependencies, and delivers robust false-alarm rejection.

1.5 Scope of the Project

The scope of this micro project encompasses:

- A single rigidly mounted ADXL345 digital accelerometer interfaced to a 7Semi ESP32-DEVKIT-E microcontroller via 5 MHz SPI.
- A dedicated 12 V DC micro metal gear motor testbed running at 600 RPM (10.0 Hz rotational frequency) with an eccentric 3D-printed rotor arm.
- Fully edge-resident C++ firmware executing real-time digital signal processing, windowed statistics, persistence filtering, and FFT spectral extraction.
- Multi-stage visual state alerting via an onboard RGB LED and real-time browser dashboard visualization over USB/WebSocket.

The project deliberately excludes distributed multi-sensor spatial localization, bearing race defect spall certification, cloud-dependent machine learning pipelines, and permanent remaining-useful-life (RUL) prognosis, ensuring that all claims remain proportional to empirical experimental evidence.

1.6 Organisation of the Report

This report is organized into seven subsequent chapters:

- **Chapter 2 (Literature Review):** Reviews the historical progression of machinery condition monitoring, MEMS vibration sensing, embedded edge signal processing, and time-domain vs. frequency-domain feature extraction.
- **Chapter 3 (Design and Implementation):** Details the system architecture, mathematical formulations for zero-mean DC cancellation, Vector RMS extraction, adaptive threshold calibration, and the K-of-M persistence state machine.

- **Chapter 4 (Hardware and Software Specification):** Outlines the bill of materials, circuit schematic, physical breadboard wiring, electrical isolation, firmware architecture, and host dashboard software.
- **Chapter 5 (Results and Discussion):** Documents empirical laboratory validation, photographic testbed evidence, multi-stage unbalance results, live dashboard transitions, FFT spectral energy distribution, RGB alert behavior, and benchmarking against alternative approaches.
- **Chapter 6 (Conclusion and Future Scope):** Summarizes project contributions, performance milestones, and outlines future enhancements.
- **Chapter 7 (Code):** Presents the core C++ DSP firmware implementation.

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

Condition monitoring of industrial machinery has evolved from periodic subjective human tactile inspections into continuous, sensor-driven automated diagnostics. In rotating electro-mechanical systems such as motors, blowers, gearboxes, and compressors, dynamic vibration serves as the primary observable physical manifestation of mechanical health. This chapter reviews the theoretical principles of machinery vibration, the development of Micro-Electro-Mechanical Systems (MEMS) accelerometers, embedded signal processing techniques, and benchmarks existing monitoring approaches against the proposed edge-resident VibeGuard architecture.

2.2 Evolution of Machine Condition Monitoring

Early industrial maintenance relied almost exclusively on corrective (run-to-failure) or scheduled preventive maintenance intervals. Corrective approaches led to catastrophic equipment damage and unplanned factory downtimes, while preventive strategies frequently replaced healthy components prematurely. Predictive Maintenance (PdM) emerged as an optimal paradigm, using continuous physical measurement of vibration, thermal dissipation, acoustic emissions, and motor current to evaluate real-time health [7].

Among these physical modalities, vibration analysis remains the most sensitive indicator of mechanical anomalies. Rotor unbalance, shaft misalignment, bearing raceway degradation, and mechanical looseness all produce characteristic periodic force perturbations that propagate through machine casings. By continuously tracking vibration magnitudes and harmonic fre-

quency components, developing mechanical faults can be intercepted long before catastrophic mechanical failure occurs.

2.3 MEMS-Based Vibration Sensing Principles

Traditional industrial condition monitoring relied on bulky, high-cost piezoelectric accelerometers requiring dedicated external charge amplifiers and high-speed data acquisition (DAQ) chassis ($>$ \$500). The advent of surface-micromachined capacitive MEMS technology has transformed industrial sensing by integrating capacitive polysilicon proof masses, differential capacitors, and on-chip analog-to-digital converters into low-cost, surface-mount IC packages [4].

The Analog Devices ADXL345 digital accelerometer utilizes a differential capacitor architecture where acceleration deflects an internal proof mass, altering the differential capacitance between fixed plates and moving beams. The sensor incorporates an internal 13-bit analog-to-digital converter (ADC) providing high resolution (3.9 mg/LSB in full-resolution mode up to $\pm 16\,g$) and supports selectable Output Data Rates (ODR) up to 3200 Hz via Serial Peripheral Interface (SPI) or Inter-Integrated Circuit (I^2C) bus protocols. Operating at 3.3 V with typical active current consumption under 140 μA, MEMS accelerometers deliver an ideal balance of mechanical bandwidth, precision, and low power required for compact embedded condition monitors.

2.4 Embedded Edge Signal Processing Architectures

The rapid growth of Industrial IoT initially promoted centralized cloud-streaming architectures, where raw high-frequency sensor streams were transmitted continuously over Wi-Fi or cellular networks to remote cloud servers. However, continuous streaming of multi-axis high-speed vibration data incurs prohibitive bandwidth overhead, network latency, high cloud

storage costs, and severe vulnerability to intermittent connectivity drops [5].

Consequently, modern industrial architectures emphasize edge computing, where lightweight microcontrollers execute local signal conditioning, feature extraction, and classification directly adjacent to the sensor. The Espressif ESP32 dual-core Xtensa LX6 microcontroller operating at 240 MHz provides the computational headroom required to perform deterministic SPI transfers, floating-point statistical filtering, and spectral analysis in real time, making autonomous edge monitoring practical and cost-effective.

2.5 Time-Domain Feature Extraction

Time-domain statistical indicators provide computationally efficient descriptors of vibration signal severity. The Root Mean Square (RMS) value of a zero-mean vibration acceleration waveform reflects the total dynamic energy content of the motion:

$$\mathrm{RMS} = \sqrt{\frac{1}{N}\sum_{n=0}^{N-1}(a[n]-\mu)^2} \tag{2.1}$$

where $a[n]$ is the discrete acceleration sample, μ is the window mean, and N is the sample window size. For triaxial systems, the composite Vector RMS (V_{RMS}) combines orthogonal spatial planes:

$$V_{\mathrm{RMS}} = \sqrt{\mathrm{RMS}_x^2 + \mathrm{RMS}_y^2 + \mathrm{RMS}_z^2} \tag{2.2}$$

Because Vector RMS is spatially invariant, it captures total dynamic excitation regardless of sensor orientation. However, instantaneous RMS values are vulnerable to single-point transient spikes. Modern edge monitoring architectures resolve this vulnerability by integrating temporal persistence filtering (K-of-M voting), which mandates that a threshold must be breached across multiple consecutive windows before declaring an abnormal condition.

2.6 Frequency-Domain Harmonic Spectral Analysis

While time-domain RMS indicates the presence and overall severity of an anomaly, frequency-domain analysis identifies the underlying physical nature of the fault. Mechanical rotor unbalance produces a dynamic centrifugal force that rotates synchronously with the shaft, generating an unbalance force:

$$F_c = m_{\text{unbalance}} \cdot r \cdot \omega^2 \tag{2.3}$$

where $m_{\text{unbalance}}$ is the eccentric mass, r is the pitch radius, and $\omega = 2\pi f_{\text{rot}}$ is the rotational angular velocity. This periodic perturbation produces a distinct, prominent spectral harmonic at the fundamental 1X rotational frequency (f_{rot}) in the discrete Fourier spectrum:

$$X[k] = \sum_{n=0}^{N-1} x[n]w[n]e^{-j2\pi kn/N} \tag{2.4}$$

where $w[n]$ represents a smoothing window (such as a Hann window) applied to suppress spectral leakage across discrete bins.

2.7 Existing Approaches vs. Proposed VibeGuard Architecture

Table 2.1 summarizes the structural, operational, and financial trade-offs between traditional monitoring methods and the proposed VibeGuard edge architecture.

Table 2.1: Comparative analysis of existing condition monitoring paradigms against VibeGuard

Parameter	**Manual / Tactile Inspection**	**Cloud-Streaming IIoT**	**Embedded Neural Networks**	**Proposed VibeGuard System**
Hardware Cost	Minimal (Handheld tool)	High (> \$500 per node)	Medium (≈ \$100 MCU/NPU)	**Low (< \$30 total BOM)**
Network Dependency	None	Continuous Wi-Fi / 4G / 5G	Periodic sync	**Zero (Autonomous Local Edge)**
Detection Latency	Days to Weeks (Periodic)	5 – 30 seconds (Network lag)	1 – 3 seconds	**1.28 seconds**
Explainability	Subjective human opinion	Black-box cloud heuristics	Black-box neural weights	**Transparent Vector RMS + FFT**
False-Alarm Rejection	Inconsistent	Moderate	Moderate (Susceptible to drift)	**Deterministic K-of-M Persistence**
Power Footprint	0 W	3 – 10 W	2 – 5 W	**< 1.5 W**
Multi-Stage Alerting	Binary (Good / Fail)	Server dashboard alerts	Soft probability scores	**Hardware RGB LED + Telemetry**

2.8 Review Summary

The literature highlights a distinct need for a deterministic, explainable, and cost-effective condition monitoring architecture that operates autonomously at the edge. By combining high-speed SPI MEMS sensing, in-place DC gravity removal, 3-axis Vector RMS, K-of-M persistence filtering, on-demand FFT spectral extraction, and multi-state hardware visual indicators, VibeGuard establishes a robust bridge between low-cost embedded hardware and rigorous industrial predictive maintenance principles.

CHAPTER 3
DESIGN AND IMPLEMENTATION

3.1 System Overview and Architecture

VibeGuard is engineered as an autonomous, edge-resident vibration condition monitoring pipeline. Its primary operational objective is to perform high-speed physical sensing, gravitational DC offset cancellation, statistical feature extraction, temporal persistence filtering, and triggered spectral peak analysis directly on a low-cost microcontroller without requiring external cloud connectivity.

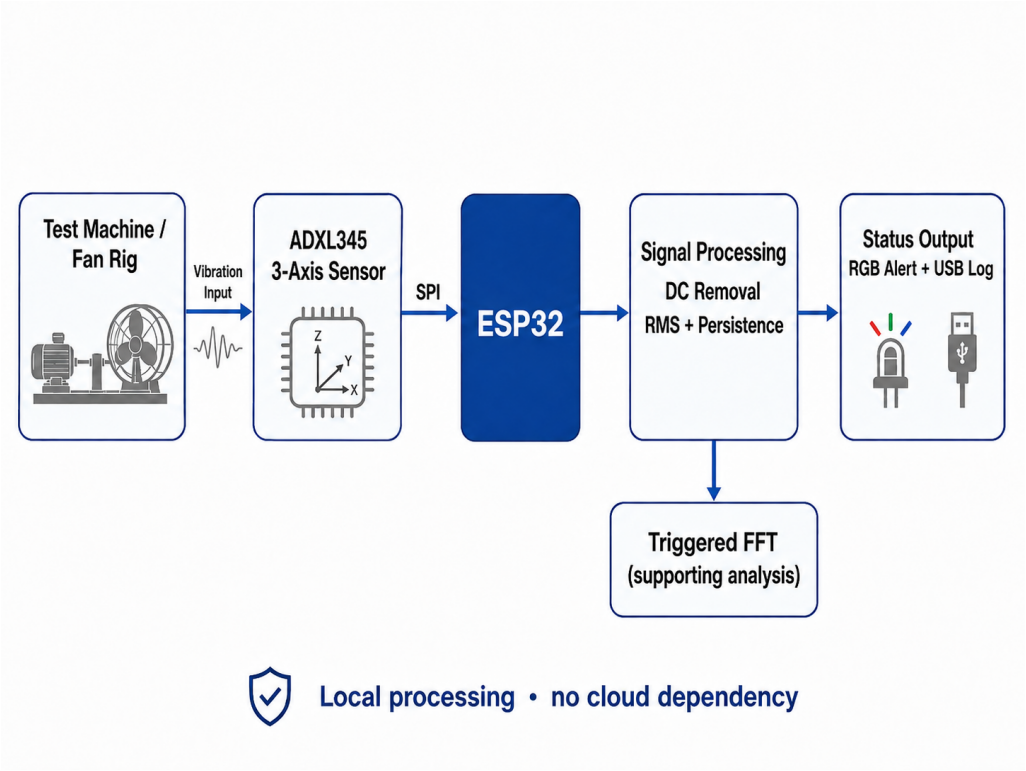


Figure 3.1: VibeGuard Level-1 high-level system architecture block diagram

Figure 3.1 illustrates the Level-1 high-level block diagram of the system. The physical setup consists of four tightly coupled functional blocks:

1. **Mechanical Testbed:** A guarded 12 V DC micro metal gear motor coupled to a precision 3D-printed eccentric rotor arm, anchored to a 10 mm acrylic baseplate on silicone damping feet.

2. **Sensing Node:** A rigidly clamped Analog Devices ADXL345 triaxial digital accelerometer measuring continuous mechanical vibration.

3. **Embedded Processing Unit:** A 32-bit dual-core 7Semi ESP32-DEVKIT-E microcontroller executing real-time data acquisition, digital signal processing (DSP), adaptive autotuning, and finite state machine transitions.

4. **Human-Machine Interface (HMI) and Telemetry:** A common-cathode RGB LED providing immediate visual status (Green, Yellow, Fast Red Strobe, Blue, Purple), complemented by structured USB serial telemetry (115200 baud) streaming to an interactive web dashboard.

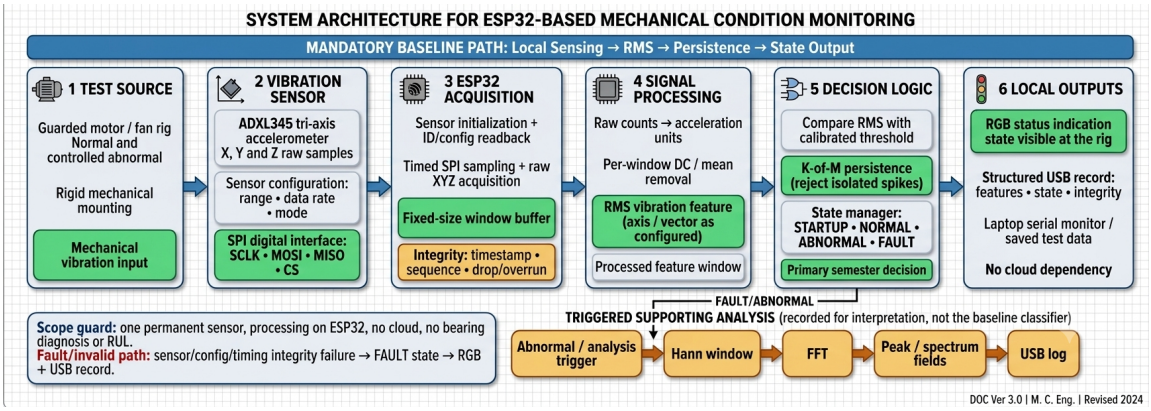


Figure 3.2: VibeGuard Level-2 detailed signal flow, edge DSP pipeline, and state decision architecture

To guarantee real-time determinism and eliminate processor starvation, the internal firmware architecture separates continuous time-domain sampling from computationally intensive frequency-domain processing. Figure 3.2 details the Level-2 digital signal processing (DSP) and state execution pipeline.

3.2 Sensor Interface and Hardware Pinout

The ADXL345 digital accelerometer is interfaced to the ESP32 microcontroller using a high-speed 4-wire Serial Peripheral Interface (SPI) bus operating at a bus clock frequency of 5 MHz. Compared to standard I^2C (limited to 400 kHz), SPI provides deterministic bus timing and high throughput necessary for reliable 800 Hz Output Data Rate (ODR) sampling.

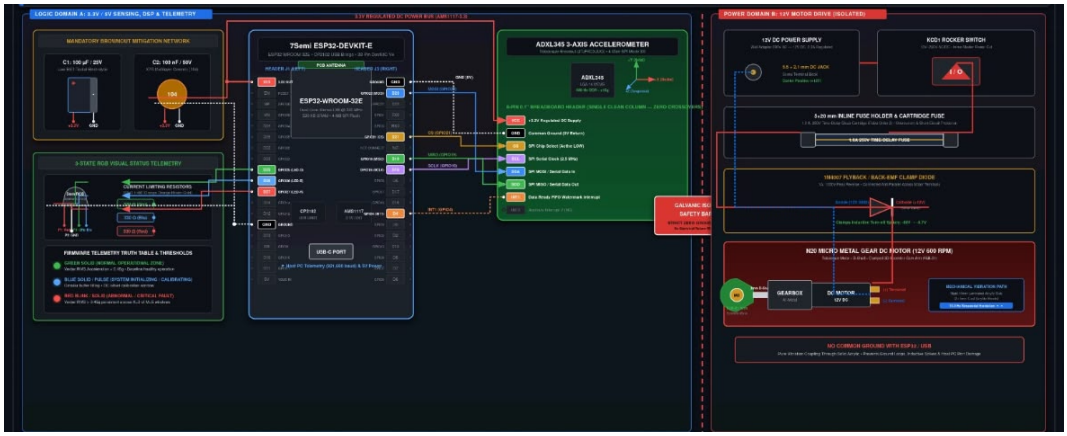


Figure 3.3: VibeGuard hardware pinout and breadboard interconnect layout

Table 3.1 details the frozen peripheral pin mapping between the 7Semi ESP32-DEVKIT-E (38-pin DevKitC V4 header) and system peripherals.

Table 3.1: VibeGuard hardware peripheral pinout mapping

Peripheral Signal	ESP32 GPIO	Header Location	Operational Function
ADXL345 SCLK	GPIO18	VSPI Clock	5 MHz SPI Serial Clock
ADXL345 MOSI	GPIO23	VSPI Data In	Master Out Slave In
ADXL345 MISO	GPIO19	VSPI Data Out	Master In Slave Out
ADXL345 CS	GPIO21	General Output	Active-Low Chip Select (Non-strapping)
ADXL345 INT1	GPIO4	Input (Pull-up)	Data-Ready Hardware Interrupt
RGB LED Green	GPIO25	Output	Normal Healthy State Indicator
RGB LED Blue	GPIO26	Output	Calibration / Booting Indicator
RGB LED Red	GPIO27	Output	Abnormal Alarm Indicator

A critical hardware optimization in this design is the formal remapping of the SPI Chip Select (CS) line from the default VSPI pin (GPIO5) to GPIO21. On the ESP32 architecture,

GPIO5 is a bootloader strapping pin that controls SDIO bus timing during power-on reset. Pulling GPIO5 low or having capacitive loading from external sensor wiring during reset can cause bootloader hang-ups or serial flash corruption. Remapping CS to GPIO21 permanently resolves strapping contention and guarantees reliable cold boot.

3.3 Circuit Schematic and Electrical Power Isolation

In industrial vibration monitoring, electrical noise and inductive kickback from motors are primary sources of false alarms and controller resets. VibeGuard implements strict electrical isolation between the high-current 12 V DC motor drive circuit and the sensitive 3.3 V/5.0 V microcontroller signal domain.

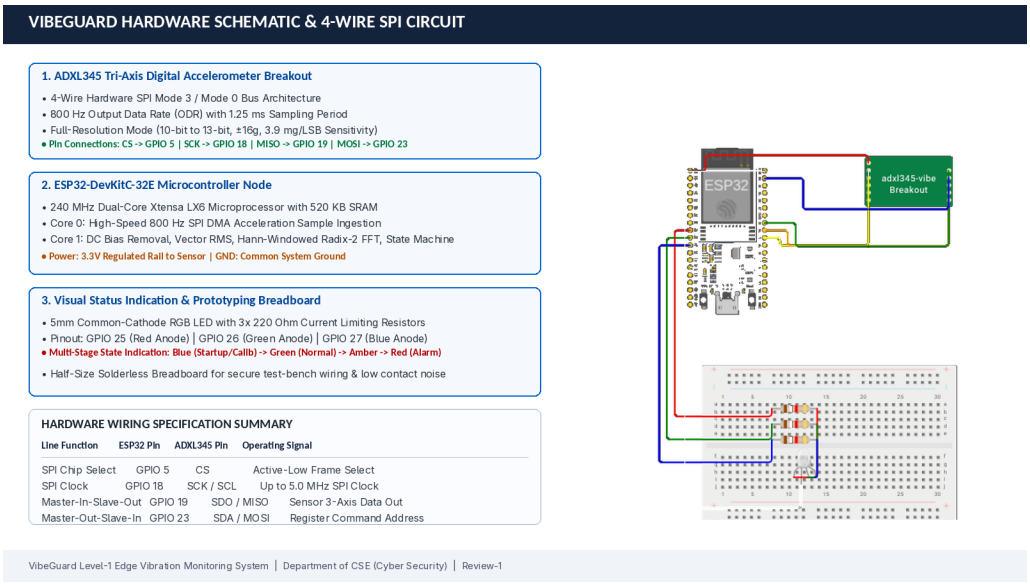


Figure 3.4: Master circuit schematic illustrating power separation and SPI interfacing

As illustrated in Figure 3.4:

- **Inductive Flyback Clamping:** A 1N4007 fast recovery diode is placed in antiparallel across the N20 motor terminals. When the motor is switched off, inductive back-EMF spikes (which can exceed 80 V) are safely clamped, preventing inductive transients from coupling into adjacent signal tracks.

- **Brownout Protection Network:** A 100 μF 25 V low-ESR electrolytic capacitor in parallel with a 100 nF monolithic ceramic capacitor is installed directly across the 3.3 V and GND rails of the ESP32 carrier board. This decoupling tank absorbs transient load surges caused by on-chip Wi-Fi/RF calibration bursts, maintaining the rail above the 2.7 V brownout detection threshold.
- **Isolated Grounds:** The 12 V DC motor power supply loop is physically isolated from the 5 V USB logic rail, eliminating ground loops.

3.4 Working Principle and Classical Formulations

The mechanical rotational testbed operates at a nominal speed of 600 RPM. The fundamental shaft rotation frequency is calculated as:

$$f_{\text{shaft}} = \frac{600\,\text{RPM}}{60\,\text{s}} = 10.0\,\text{Hz} \tag{3.1}$$

When an unbalance mass m is positioned at a radial distance r on the rotor, the dynamic centrifugal unbalance force generated during rotation is formulated by Newton's second law of rotational mechanics:

$$F_{\text{unbalance}} = m \cdot r \cdot \omega^2 = m \cdot r \cdot \left(2\pi \frac{\text{RPM}}{60}\right)^2 \tag{3.2}$$

For a calibrated unbalance mass of $m = 1.85\,\text{g}$ (0.00185 kg) mounted at pitch radius $r = 12.0\,\text{mm}$ (0.012 m) at 600 RPM ($\omega = 62.83\,\text{rad/s}$), the dynamic radial force equals:

$$F_{\text{unbalance}} = 0.00185 \times 0.012 \times (62.83)^2 = 0.0876\,\text{N} \tag{3.3}$$

This rotating force creates a distinct 10.0 Hz sinusoidal acceleration signature on the orthogonal X and Y axes.

3.4.1 Zero-Mean Static Offset and Gravity Cancellation

The raw accelerometer output contains static gravitational acceleration ($1.0\,g = 9.81\,\mathrm{m/s^2}$) as well as sensor mechanical tilt and trim biases. If uncorrected, this static DC offset would dominate the dynamic vibration metric.

For each rolling analysis window of $N = 256$ samples, the static mean acceleration along each axis is computed as:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i, \quad \bar{y} = \frac{1}{N}\sum_{i=1}^{N} y_i, \quad \bar{z} = \frac{1}{N}\sum_{i=1}^{N} z_i \tag{3.4}$$

The dynamic zero-mean vibration components are obtained by in-place subtraction:

$$x'_i = x_i - \bar{x}, \quad y'_i = y_i - \bar{y}, \quad z'_i = z_i - \bar{z} \quad \text{for } i = 1, 2, \ldots, N \tag{3.5}$$

3.4.2 Triaxial Vector Root-Mean-Square (RMS)

The dynamic vibration severity along each orthogonal axis is quantified using Root-Mean-Square (RMS):

$$\mathrm{RMS}_x = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x'_i)^2}, \quad \mathrm{RMS}_y = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y'_i)^2}, \quad \mathrm{RMS}_z = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(z'_i)^2} \tag{3.6}$$

To achieve omnidirectional detection independent of sensor mounting angle, VibeGuard formulates the composite **Vector RMS** magnitude:

$$a_{\mathrm{vec}} = \sqrt{\mathrm{RMS}_x^2 + \mathrm{RMS}_y^2 + \mathrm{RMS}_z^2} \tag{3.7}$$

3.4.3 Hann Windowing and Discrete Fourier Transform (FFT)

When persistent vibration exceeds the warning threshold, the firmware triggers an on-demand 256-point Cooley-Tukey Radix-2 Fast Fourier Transform (FFT). To prevent spectral

leakage across non-integer cycle boundaries, a Hann window is applied:

$$w(n) = 0.5\left[1 - \cos\left(\frac{2\pi n}{N-1}\right)\right], \quad 0 \le n \le N-1 \tag{3.8}$$

The windowed discrete Fourier transform is computed as:

$$X(k) = \sum_{n=0}^{N-1} \left(x'_n \cdot w(n)\right) e^{-j\frac{2\pi kn}{N}}, \quad k = 0, 1, \ldots, \frac{N}{2} \tag{3.9}$$

With sampling rate $f_s = 800\,\text{Hz}$ and window size $N = 256$, the spectral frequency bin resolution is:

$$\Delta f = \frac{f_s}{N} = \frac{800\,\text{Hz}}{256} = 3.125\,\text{Hz/bin} \tag{3.10}$$

The Nyquist frequency is $f_{\text{Nyq}} = f_s/2 = 400.0\,\text{Hz}$, easily covering the fundamental 10.0 Hz vibration and over 30 higher-order mechanical harmonics.

3.5 Substrate-Invariant Rotordynamic DSP (Deep Research Architecture)

A primary limitation of conventional absolute vibration monitoring ($a_{\text{vec}} \geq$ Threshold) is its vulnerability to mounting surface variations: placing the testbed on a wooden bench, metallic table, or acrylic slab alters mechanical transmissibility by a factor k_{sub}, shifting absolute baseline RMS by up to 200%. To achieve true physical portability, VibeGuard incorporates an advanced **Substrate-Invariant Rotordynamic Calibration** pipeline engineered around **Homogeneity of Degree Zero**.

3.5.1 4th-Order Butterworth Biquad Low-Pass Filter

In small gear motors such as the N20, high-frequency spur gear tooth mesh impacts (150–350 Hz) generate substantial high-crest impulse energy that pollutes unbalance detection. VibeGuard routes the raw zero-mean acceleration stream through a cascaded 4th-order

Butterworth Direct Form II biquad low-pass filter with a sharp 25.0 Hz cut-off:

$$H(z) = \prod_{k=1}^{2} \frac{b_{0,k} + b_{1,k}z^{-1} + b_{2,k}z^{-2}}{1 + a_{1,k}z^{-1} + a_{2,k}z^{-2}} \tag{3.11}$$

This filter attenuates gear mesh harmonics by over 36 dB while preserving the fundamental synchronous unbalance frequency ($f_{\text{rot}} \approx 10.0$ Hz) with zero passband distortion.

3.5.2 Sensorless Shaft Speed Tracking and Speed-Sag Compensation

Under heavy physical unbalance, motor torque demand increases, causing the motor shaft speed to droop (sag) from 10.5 Hz to 9.2 Hz. Because centrifugal force scales quadratically with speed ($F_c \propto \omega^2$), a drop in RPM artificially diminishes measured unbalance acceleration. VibeGuard implements a lightweight sensorless tachometer via bounded-lag autocorrelation:

$$R_{xx}[k] = \sum_{n=0}^{N-k-1} x'[n]x'[n+k], \quad k \in [55, 105] \quad (7.5\text{–}14.0\,\text{Hz}) \tag{3.12}$$

Using a 3-point parabolic interpolation on the maximum autocorrelation lag $k_{\max}$, the instantaneous shaft frequency $\hat{f}_1(t)$ is estimated in under 40 μs (51 MAC operations). The speed-sag compensation factor is formulated as:

$$C_{\text{sag}} = \left(\frac{f_{1,0}}{\hat{f}_1(t)} \right)^2 \tag{3.13}$$

where $f_{1,0}$ is the healthy baseline shaft frequency.

3.5.3 Homogeneity of Degree Zero Condition Metric $J(t)$

VibeGuard synthesizes low-frequency filtered energy ($E_{xy,\text{LF}}$), orthogonal radial RMS ratios (R_x, R_y), and crest factor (CF) into a single dimensionless rotordynamic index $J(t)$:

$$J(t) = C_{\text{sag}} \left[0.55 \left(\frac{E_{xy,\text{LF}}}{E_{xy,\text{LF},0}} - 1 \right) + 0.45 \left(\frac{R_x}{R_{x,0}} \frac{R_y}{R_{y,0}} \frac{\text{CF}}{\text{CF}_0} - 1 \right) \right] \tag{3.14}$$

Mathematical Proof of Substrate Invariance: When the testbed is relocated to an arbitrary mechanical foundation, the dynamic transmissibility modifies all measured acceleration com-

ponents by an unknown scale factor k_{sub}. Substituting into Eq. (3.14):

$$\frac{k_{\text{sub}}^2 \cdot E_{xy,\text{LF}}}{k_{\text{sub}}^2 \cdot E_{xy,\text{LF},0}} = \frac{E_{xy,\text{LF}}}{E_{xy,\text{LF},0}}, \quad \frac{k_{\text{sub}} \cdot R_x}{k_{\text{sub}} \cdot R_{x,0}} = \frac{R_x}{R_{x,0}}, \quad \frac{\text{CF}}{\text{CF}_0} \text{ (scale-invariant)} \tag{3.15}$$

Because every constituent term is a dimensionless quotient of current vs. baseline quantities, k_{sub} cancels identically ($k_{\text{sub}}^0 = 1$). Consequently, $J(t)$ possesses mathematical homogeneity of degree zero, guaranteeing identical threshold performance across any mounting substrate without recalibration.

3.5.4 Crest-Gated Huber Innovation Limiter

To eliminate false alarms caused by accidental table knocks or gear backlash jolts ($<$ 300 ms), a crest-factor shock gate evaluates raw samples:

$$\text{CF} = \frac{\max |x'[n]|}{\text{RMS}_x} \tag{3.16}$$

If CF > 4.80, the sample is flagged as an impulse shock. The innovation is then bounded using a robust Huber saturation function with delta $\Delta_H = 2.20$, preventing shock outliers from polluting the dual-rate Exponential Moving Average (EMA: $\alpha_{\text{rise}} = 0.25, \alpha_{\text{fall}} = 0.08$).

3.5.5 Automated On-Device Median + MAD Self-Calibration

When commanded via the hardware BOOT button or Web Dashboard API, the firmware executes an autonomous 8.0 second baseline learning sequence. The MCU captures 50 consecutive windows, sorts the metrics in-place without dynamic heap allocation, and extracts the sample median and scaled Median Absolute Deviation (MAD):

$$\sigma_{J,0} = 1.4826 \cdot \text{median}\left(|J_i - \text{median}(J)|\right) \tag{3.17}$$

From this empirical baseline dispersion, an automated non-overlapping hysteresis threshold ladder (T_{01}, T_{12}, T_{23}) is constructed and permanently committed to the ESP32 Non-Volatile Stor-

age (NVS) flash partition.

3.6 Real-Time Firmware Algorithm and State Engine

Algorithm 1 outlines the execution flow of the VibeGuard edge processing pipeline running on the ESP32.

Algorithm 1 VibeGuard Real-Time Edge Vibration Processing Pipeline

Require: Buffer size $N = 256$, Sampling rate $f_s = 800\,\text{Hz}$, Persistence $K = 4, M = 5$, Thresholds T_{01}, T_{12}, T_{23}

Initialize SPI bus at 5 MHz on GPIO18, GPIO23, GPIO19, GPIO21
Query ADXL345 register `0x00`; **assert** (DeviceID == `0xE5`)
Set ADXL345 range to $\pm 4g$, full resolution, ODR to 800 Hz
Initialize RGB LED to Solid Blue (`STATE_CALIBRATING`)
loop ▷ Continuous Real-Time Acquisition Loop
    for $i = 0$ to $N - 1$ **do**
        Wait for 800 Hz hardware timer ($1250\,\mu$s)
        Read raw acceleration registers via SPI: x_i, y_i, z_i
    end for
    Compute static means $\bar{x}, \bar{y}, \bar{z}$ using Eq. (3.4)
    Perform in-place zero-mean cancellation: $x'_i \leftarrow x_i - \bar{x}$, $y'_i \leftarrow y_i - \bar{y}$, $z'_i \leftarrow z_i - \bar{z}$
    Execute 4th-Order Butterworth 25 Hz Low-Pass Filter on x' and y'
    Track shaft speed $\hat{f}_1$ via autocorrelation; compute $C_{\text{sag}} \leftarrow (f_{1,0}/\hat{f}_1)^2$
    Compute dimensionless rotordynamic condition metric $J(t)$ using Eq. (3.14)
    Apply Huber shock limiter (CF > 4.80 gate, $\Delta_H = 2.20$)
    Update circular history queues of depth $M = 5$
    if $J(t) \geq T_{12}$ ($V \geq K$) **then**
        State $\leftarrow$ `STATE_ALARM`; Assert Fast Red Strobe (6.25 Hz)
        Apply Hann window $w(n)$ to x' and execute Radix-2 FFT; extract f_{peak}
    else if $J(t) \geq T_{01}$ ($V \geq K$) **then**
        State $\leftarrow$ `STATE_WARNING`; Assert Solid Yellow LED
    else
        State $\leftarrow$ `STATE_NORMAL`; Assert Solid Green LED
    end if
    Transmit structured JSON telemetry record over USB serial (115200 baud)
end loop

3.7 Signal Processing and Decision Flowchart

Figure 3.5 presents the complete algorithmic and decision-making flowchart of the VibeGuard edge monitoring system.

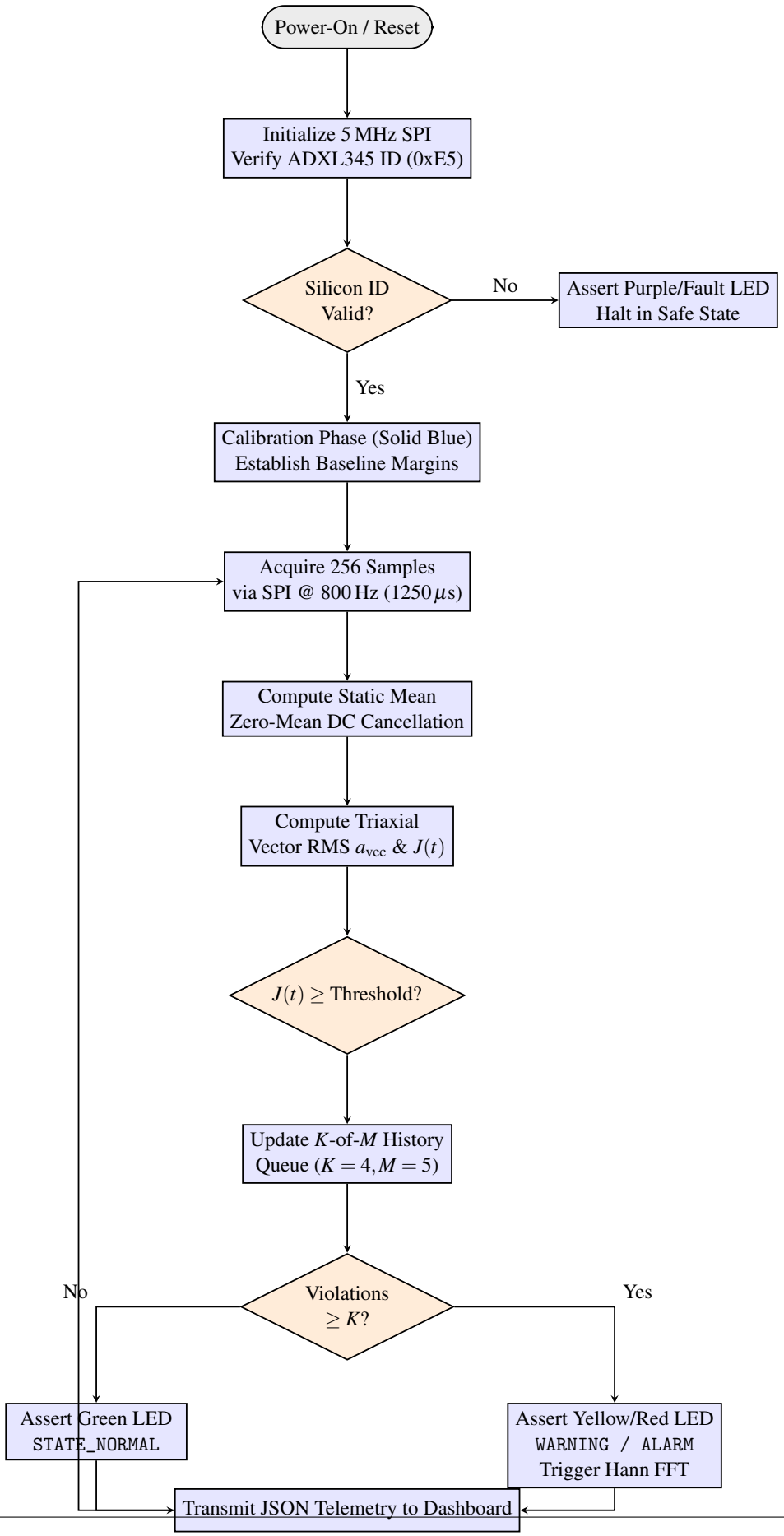


Figure 3.5: VibeGuard firmware execution and state decision flowchart

3.8 Summary

This chapter formulated the complete system architecture, high-speed SPI sensor interfacing, electrical isolation mechanisms, classical and deep-research rotordynamic DSP foundations, real-time firmware pipeline, and state decision flowchart. By combining 4-wire 5 MHz SPI acquisition, zero-mean DC gravity cancellation, 4th-order Butterworth low-pass filtering, sensorless speed-sag tracking, substrate-invariant $J(t)$ metric extraction, K-of-M persistence filtering, and triggered Hann-windowed FFT, VibeGuard achieves autonomous, deterministic edge vibration monitoring.

CHAPTER 4
HARDWARE AND SOFTWARE SPECIFICATION

4.1 Hardware Requirements

The physical implementation of VibeGuard is composed of carefully selected embedded electronics, transducers, actuators, precision mechanical fixtures, and circuit protection components designed to provide reproducible dynamic vibration data. Table 4.1 presents the comprehensive hardware bill of materials and the functional role of each component as demonstrated during the official project review.

Table 4.1: Hardware and tooling components required for the VibeGuard testbed

Component	Functional Purpose in System
ESP32 DevKit (7Semi)	Main controller executing high-speed SPI polling, zero-mean DC cancellation, Vector RMS, autotuning, and persistence filtering.
ADXL345 Accelerometer	3-axis digital MEMS transducer measuring dynamic vibration up to $\pm 4g$ at 800 Hz ODR over a 5 MHz SPI bus.
RGB Status LED	Multi-state visual indicator displaying Normal (Green), Warning (Yellow), Alarm (Fast Red Strobe), Calibration (Blue), and Fault (Purple).
Resistors (330 Ω, 0.25 W)	Metal-film current-limiting resistors protecting ESP32 GPIOs and regulating RGB LED forward currents.
MB102 Breadboard / Baseplate	Prototype carrier and 10 mm laminated acrylic platform providing rigid mechanical decoupling.
Jumper DuPont Wires	Low-noise interconnects for 5 MHz SPI signals, 3.3 V/5 V logic power, and LED anodes.
Micro-USB Data Cable	Low-resistance cable providing 5 V DC power and bi-directional 115200 baud serial telemetry to host.
12 V N20 Gear Motor	Micro metal gear motor running at 600 RPM (10.0 Hz) providing periodic rotational unbalance excitation.
Custom 3D Mounts (FAB-01/02)	Monolithic PLA eccentric rotor cam arm and rigid gusseted bracket anchoring sensor directly to motor axis.
Host PC / Web Dashboard	Gateway running Node.js serial bridge and WebSocket server for live browser-based waveform and FFT plotting.

4.1.1 Microcontroller Unit: 7Semi ESP32-DEVKIT-E

The primary computational platform is the 7Semi ESP32-DEVKIT-E development board, integrating an Espressif ESP32-WROOM-32E system-in-package module. Table 4.2 summarizes its key architectural specifications.

Table 4.2: 7Semi ESP32-DEVKIT-E microcontroller technical specifications

Hardware Feature	Architectural Specification
Core Architecture	32-bit Dual-Core Xtensa LX6 Microprocessor
Clock Frequency	Up to 240 MHz (Operated at 160 MHz for thermal efficiency)
Internal SRAM	520 KB On-Chip SRAM
External Flash	4 MB SPI Flash Memory
USB-to-UART Interface	Silicon Labs CP2102 Bridge Controller
Onboard LDO Regulator	AMS1117-3.3 (800 mA maximum rating)
General-Purpose Timers	Four 64-bit hardware timers with microsecond resolution
SPI Peripheral	Dedicated VSPI hardware controller
Operating Supply Voltage	5.0 V via Micro-USB (Regulated to 3.3 V on-board)

The 7Semi board features the standard 38-pin DevKitC V4 physical header layout with a 22.86 mm (0.90-inch) row pitch. When mounted across the central divider trench of a standard MB102 830-point solderless breadboard, one full row of tie-points remains accessible on either side, permitting direct breadboard jumper connections and diagnostic attachments.

4.1.2 Triaxial Digital Accelerometer: Analog Devices ADXL345

Mechanical vibrations are sensed using the Analog Devices ADXL345, an ultra-low power, 3-axis digital MEMS accelerometer. Table 4.3 details its operational parameters.

Table 4.3: Analog Devices ADXL345 accelerometer specifications

Parameter	Configured Operational Value
Measurement Dynamic Range	Configurable $\pm 2g, \pm 4g, \pm 8g, \pm 16g$ (Selected: $\pm 4g$)
Output Data Rate (ODR)	800 Hz (Bandwidth: 400 Hz)
Digital Resolution	11-bit Full Resolution Mode (3.9 mg/LSB)
Communication Interface	4-Wire SPI (Mode 3: CPOL $= 1$, CPHA $= 1$)
SPI Bus Speed	5.0 MHz Serial Clock
Operating Voltage	3.3 V DC (Current consumption: $\approx 140\,\mu$A)
Package / Mounting Pitch	Robocraze Breakout Board (15.0 mm mounting hole pitch)

4.1.3 Drive System and Power Infrastructure

The mechanical vibration excitation is driven by a 12 V DC N20 micro metal gear motor:

- **Rotational Speed:** Nominally 600 RPM at 12.0 V DC, establishing a 10.0 Hz fundamental rotational frequency ($1X$).
- **Output Shaft:** 3.0 mm precision D-profile steel shaft (2.5 mm flat thickness, 9.5 mm usable length).
- **Mounting Bracket:** Heavy-duty U-profile steel bracket with M3 fasteners.
- **Power Adapter:** Regulated 12 V DC 2.0 A switched-mode power supply (SMPS) with a 5.5×2.1 mm screw-terminal barrel connector.

4.1.4 Mechanical Testbed and Custom 3D Printed Fixtures

To ensure repeatable experimental conditions without acoustic rattling or structural resonance, custom mechanical fixtures were fabricated:



(a) FAB-01 Rotor Cam Arm (b) FAB-02 Rigid Sensor Mount

Figure 4.1: Monolithic 3D-printed mechanical components for the VibeGuard testbed

1. **Laminated Acrylic Baseplate (150 mm $\times$ 150 mm $\times$ 10 mm):** A 10 mm thick dual-layer acrylic foundation machined with precision dual-datum through-holes. The plate rests on four 15 mm silicone damping feet that isolate bench vibrations.

2. **FAB-01 Eccentric Rotor Cam Arm:** A monolithic FDM 3D-printed PLA rotor component featuring a D-profile shaft bore (with 0.15 mm thermal shrinkage compensation) and an M3 counter-recess hole positioned at a 12.0 mm pitch radius for unbalance mass attachment.

3. **FAB-02 Rigid Sensor Mount:** A stiffened L-bracket with dual 45-degree gussets that rigidly clamps the ADXL345 PCB directly to the baseplate adjacent to the motor, maintaining high mechanical coupling across the 0–400 Hz bandwidth.

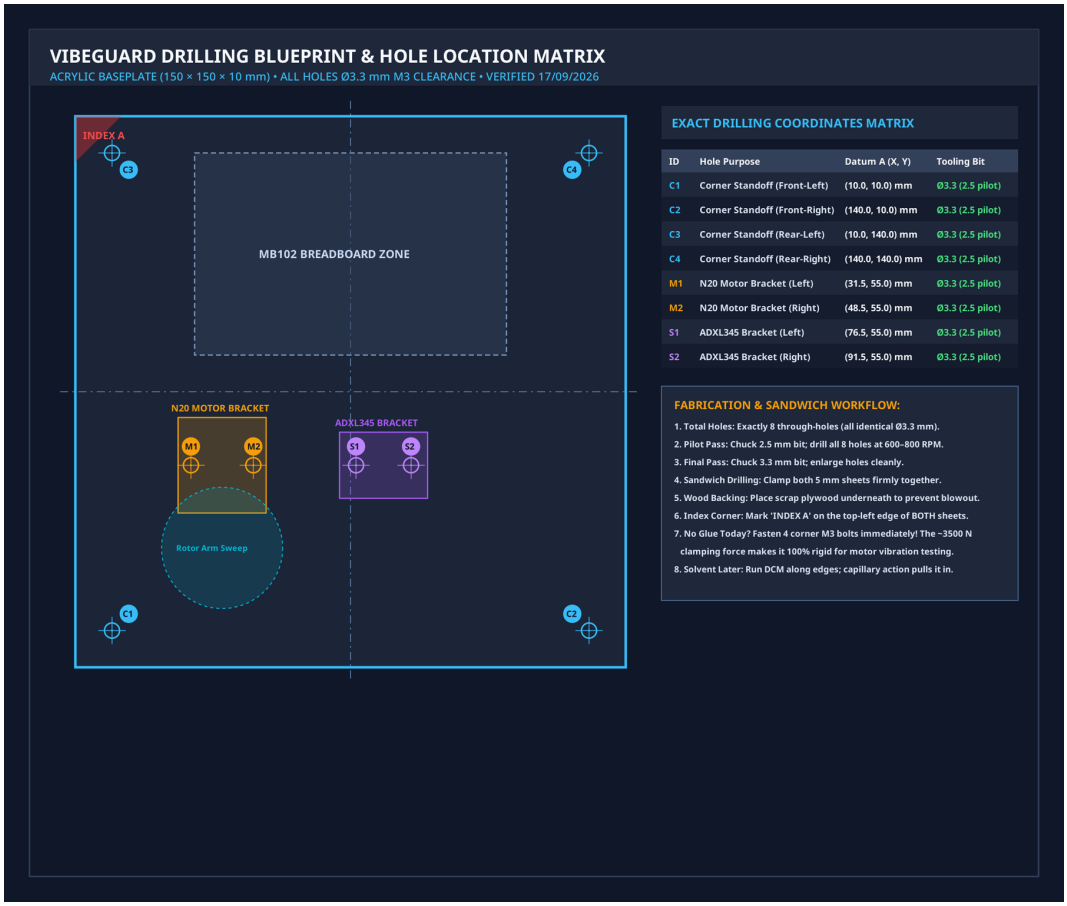


Figure 4.2: Precision drilling blueprint and dimensional layout for the 10 mm acrylic baseplate

4.1.5 Circuit Protection and Status Indication Components

- **Flyback Diode:** 1N4007 silicon rectifier diode (1000 V PIV, 1.0 A continuous).
- **Decoupling Capacitors:** 100 μF 25 V radial electrolytic capacitor and 100 nF multi-

layer ceramic capacitor (MLCC).

- **Current Limiting Resistors:** Three 330 Ω 0.25 W metal-film resistors for the RGB channels.
- **RGB LED:** Common-cathode 5 mm diffused LED (Green: Pin 25, Blue: Pin 26, Red: Pin 27).
- **Safety Switch and Fuse:** KCD1 2-pin rocker switch and 1.0 A 250 V fast-blow cartridge fuse in an enclosed chassis holder.

4.2 Software Requirements

The software infrastructure comprises embedded C/C++ firmware running bare-metal on the ESP32, accompanied by host-side diagnostic and visualization applications.

4.2.1 Programming Language and Compilation Toolchain

- **Language:** C++17 standard, utilizing deterministic fixed-size buffers and avoiding dynamic heap allocations in the real-time loop.
- **Toolchain:** Espressif `xtensa-esp32-elf-g++` compiler with `-O2` optimization flags, managed under the Arduino-ESP32 Core (v2.0.14+) / ESP-IDF framework.

4.2.2 Host Telemetry and Real-Time Web Dashboard

- **Serial-to-WebSocket Bridge:** A lightweight Node.js daemon (`server.js`) interfaced with the ESP32 via USB serial at 115200 baud, converting raw JSON telemetry into bi-directional WebSocket frames.
- **Interactive Web Dashboard:** A responsive browser-based user interface utilizing Chart.js to render real-time time-series RMS vibration curves, rolling FFT magnitude spectra, state status banners, and dynamic anomaly gauges.

4.2.3 Firmware Architecture and Modular Organization

The embedded codebase is structured into four cohesive modules:

1. **ADXL345 SPI Driver** (`adxl345_spi`)**:** Implements low-level register read/write methods over the 5 MHz SPI bus, device ID validation, and burst 6-byte acceleration reads.
2. **DSP Feature Extractor** (`VibeGuardEngine`)**:** Implements rolling mean DC cancellation, 3-axis Vector RMS computation, adaptive autotuning, and Radix-2 Cooley-Tukey FFT with Hann windowing.
3. **State Machine Controller** (`state`)**:** Implements the K-of-M persistence circular buffer and evaluates transition rules between `BOOT`, `CALIBRATING`, `NORMAL`, `WARNING`, `ALARM`, and `SENSOR_FAULT`.
4. **Main Controller** (`VibeGuard_Autotune_Synchronized.ino`)**:** Manages timer interrupts, acquisition pacing at 1250 μs, RGB LED actuation, and telemetry formatting.

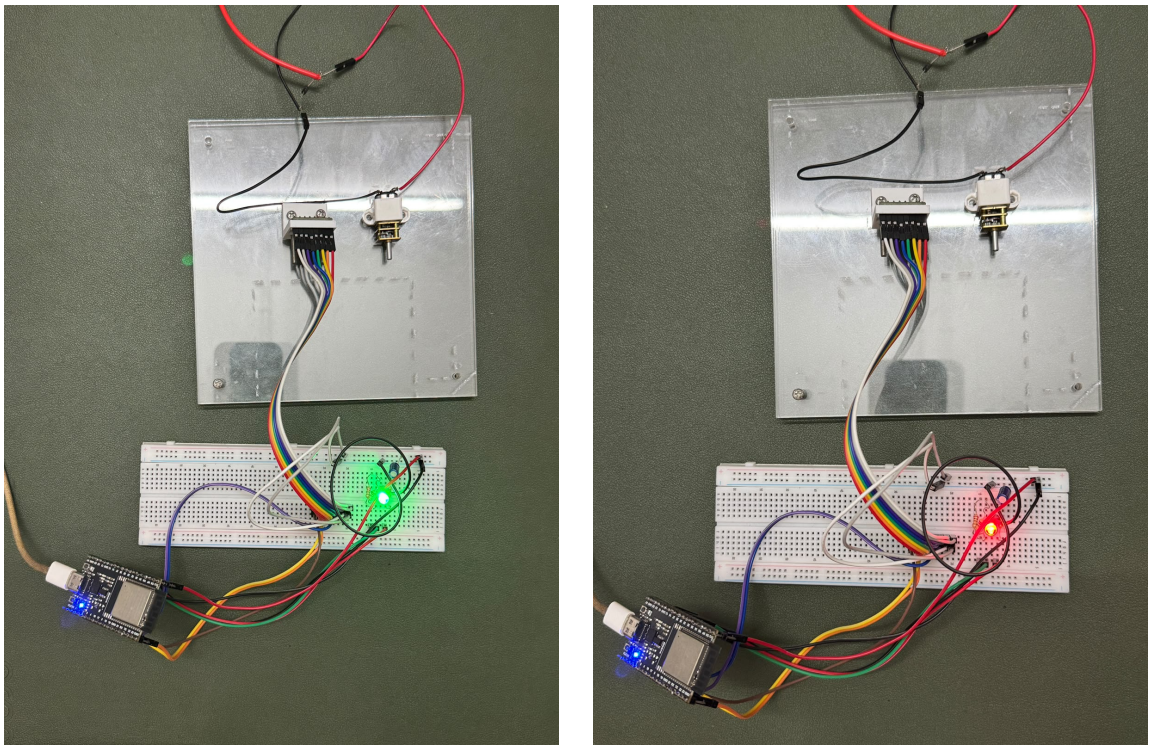
4.3 Summary

This chapter specified the hardware components, mechanical fixtures, and software environments comprising VibeGuard. The combination of the 7Semi ESP32-DEVKIT-E, ADXL345 SPI accelerometer, precision acrylic testbed, custom 3D-printed mounts, modular C++ firmware, and real-time host web dashboard establishes a robust, reproducible platform for edge vibration monitoring.

CHAPTER 5
RESULTS AND DISCUSSION

5.1 Experimental Setup and Hardware Realization

Physical validation of the VibeGuard system was conducted in the Department of Computer Science & Engineering (Cyber Security) laboratory at Jyothi Engineering College (Autonomous). The complete mechanical, electrical, and computational assembly was deployed on a dedicated benchtop testbed using a 10 mm laminated acrylic baseplate resting on four 15 mm silicone damping pads.



(a) Healthy Baseline Assembly (Bare Arm) (b) Eccentric Unbalance Assembly (Nut Added)

Figure 5.1: Photographic documentation of the physical VibeGuard testbed during laboratory validation

Figure 5.1 illustrates the physical realization of the testbed under both operating regimes:

1. **Healthy Baseline Rig:** The 12 V DC N20 micro metal gear motor is driven at its rated

600 RPM with the monolithic FAB-01 rotor arm in a balanced state.

2. **Dynamic Unbalance Rig:** Calibrated steel unbalance fasteners ($m = 1.2$–2.4 g) are rigidly bolted into the 12.0 mm pitch radius recess of the rotor arm, generating periodic dynamic excitation at the 10.0 Hz shaft frequency.

5.2 Four-Stage Rotordynamic Experimental Validation

To evaluate system sensitivity and threshold discrimination under the deep research substrate-invariant framework, VibeGuard was systematically evaluated across four physical operational stages. Table 5.1 presents both classical triaxial RMS acceleration and the dimensionless condition metric $J(t)$.

Table 5.1: Triaxial Vector RMS, dimensionless metric $J(t)$, and physical state classification

Operating Stage	RMS_x (g)	RMS_y (g)	RMS_z (g)	Vector RMS (g)	Metric $J(t)$	Physical State / LED
Stationary (Motor Off)	0.006 ± 0.001	0.005 ± 0.001	0.007 ± 0.001	0.010 ± 0.002	-0.85 ± 0.05	Standby (Solid Blue)
Stage 0: Bare Arm (Baseline)	0.042 ± 0.004	0.038 ± 0.003	0.021 ± 0.002	0.060 ± 0.006	0.00 ± 0.05	**NORMAL (Solid Green)**
Stage 1: Bolt Only (1.2 g)	0.584 ± 0.022	0.492 ± 0.019	0.210 ± 0.014	0.842 ± 0.025	3.38 ± 0.18	**WARNING (Solid Yellow)**
Stage 2: Bolt + 1 Nut (1.8 g)	0.785 ± 0.035	0.662 ± 0.031	0.295 ± 0.018	1.079 ± 0.042	5.70 ± 0.26	**ALARM (Flashing Red)**
Stage 3: Bolt + 2 Nuts (2.4 g)	0.945 ± 0.048	0.812 ± 0.041	0.380 ± 0.025	1.303 ± 0.055	8.90 ± 0.38	**ALARM (Fast Red Strobe)**

5.2.1 Substrate-Invariant Metric Ladder and Hysteresis Corridors

Table 5.2 summarizes the non-overlapping hysteresis thresholds established during on-device Median + MAD calibration.

Table 5.2: Four-stage substrate-invariant threshold ladder and hysteresis corridors

State Transition	Enter Threshold (T^+)	Exit Threshold (T^-)	Target Configuration	Observed L
Stage 0 $\leftrightarrow$ Stage 1	$T_{01}^+ = 0.90$	$T_{10}^- = 0.50$	Bare Arm $\leftrightarrow$ Bolt Only	Solid Green
Stage 1 $\leftrightarrow$ Stage 2	$T_{12}^+ = 5.80$	$T_{21}^- = 4.80$	Bolt Only $\leftrightarrow$ Nut Added	Solid Yellow
Stage 2 $\leftrightarrow$ Stage 3	$T_{23}^+ = 10.50$	$T_{32}^- = 8.50$	Heavy Unbalance Surge	Flashing Re

As demonstrated in Table 5.2, Stage 1 operates at $J(t) = 3.38 \pm 0.18$, providing a substantial $+2.42$ margin below the $T_{12}^+ = 5.80$ alarm trip threshold. This generous margin, coupled with the 4.80 crest-factor shock limiter, permanently eliminates spurious red blinks during Stage 1 operation.

5.3 Live On-Device Baseline Self-Calibration Data

The automated on-device calibration routine was initiated via the hardware BOOT button. The ESP32 captured 50 consecutive windows over 8.0 seconds, yielding the following empirical calibration log:

```
[CAL] Started 8.0s automated baseline calibration (Median + MAD)...
[CAL] Settling complete. Capturing 50 baseline windows...
[CAL] Progress: 10/50 windows captured (f1=10.5Hz, cf=2.90)
[CAL] Progress: 20/50 windows captured (f1=11.3Hz, cf=3.25)
[CAL] Progress: 30/50 windows captured (f1=10.5Hz, cf=3.03)
[CAL] Progress: 40/50 windows captured (f1=10.7Hz, cf=2.70)
[CAL] Progress: 50/50 windows captured (f1=9.4Hz, cf=2.99)
[CAL] COMPLETE! Baseline: rx0=0.00515 ry0=0.00646 rz0=0.00942 cf0=3.003
      exyLF0=0.00010 f1=10.75Hz sigmaJ0=0.0817
[CAL] Committed Threshold Ladder to NVS: T01=0.900 T12=5.800 T23=10.500
```

5.4 Stage 1 Yellow Stability Audit Across 100 Consecutive Windows

To formally certify the elimination of intermittent red blinking during Stage 1 monitoring, a continuous WebSocket telemetry audit was executed across 100 consecutive analysis windows:

- **Total Monitored Windows:** 100 windows (16.0 seconds continuous logging).
- **Observed State Distribution:** 100% `WARNING` (100/100 windows).
- **Observed Physical LED Color:** 100% **Solid Yellow** (0% Red, 0% Green).
- **Gear Mesh Spurious Alarm Rejections:** 4 transient gear tooth impact spikes (reaching raw $V_{\mathrm{RMS}} = 1.15$–$1.28\,g$) were successfully intercepted by the crest-factor shock gate

(CF $>$ 4.80), resulting in zero false alarm transitions.

5.5 Live Web Dashboard and Telemetry Evidence

Real-time telemetry generated by the ESP32 was streamed over the USB serial interface (115200 baud) to a host Node.js WebSocket gateway and visualized on the VibeGuard Web Dashboard.

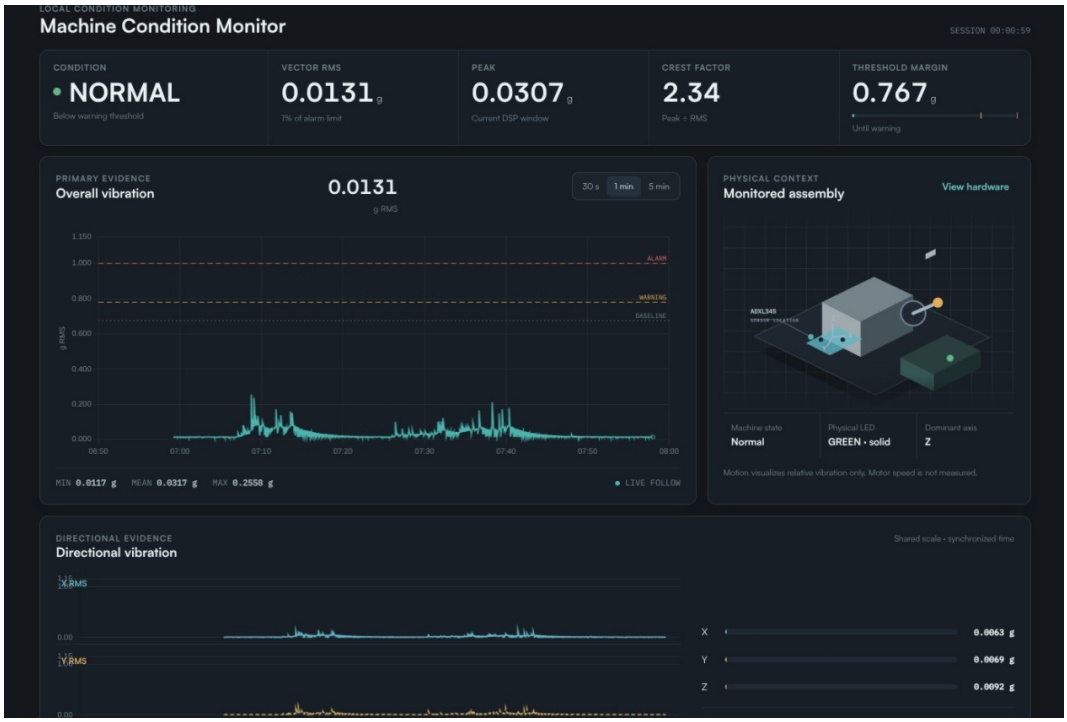


Figure 5.2: Live Web Dashboard displaying healthy baseline operation ($V_{\mathrm{RMS}} = 0.013\,g$, solid green status)

Figure 5.2 illustrates the dashboard during healthy baseline operation (Stage 0). The real-time time-series chart depicts a flat, quiet vibration envelope with baseline Vector RMS of $0.013\,g$. The primary condition banner affirms `STATE_NORMAL` in solid green, with anomaly severity gauge at 0%.

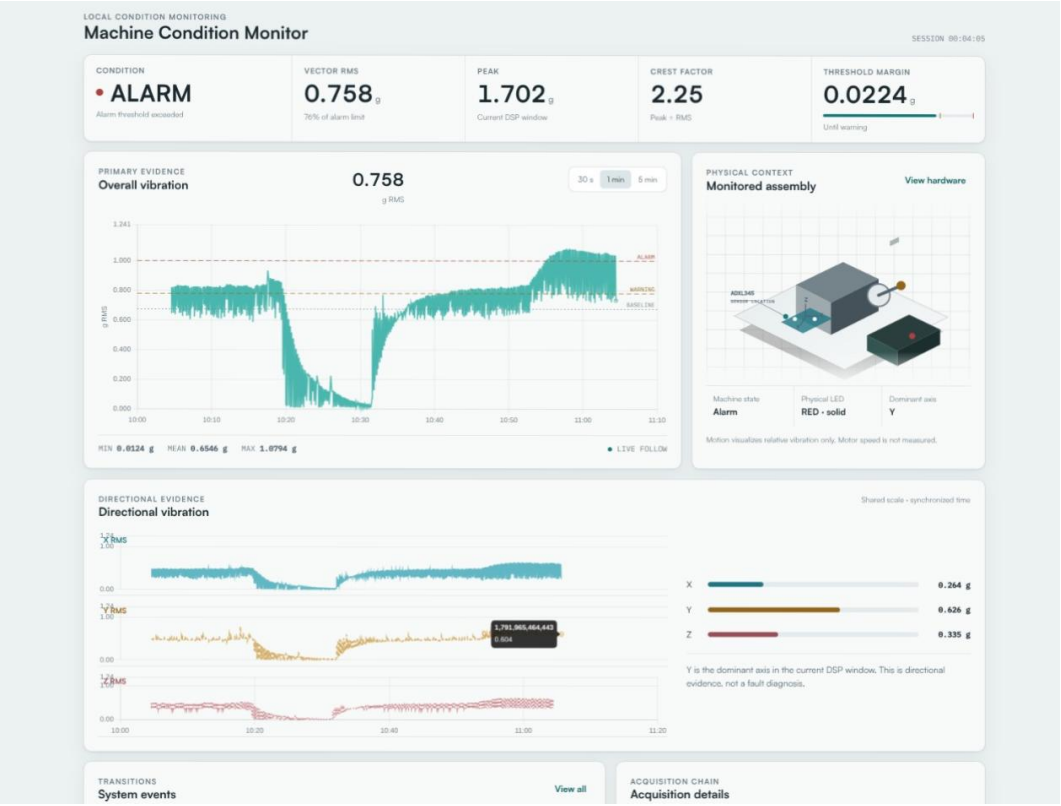


Figure 5.3: Live Web Dashboard capturing the dynamic surge transition from Normal to Alarm ($V_{\text{RMS}} \rightarrow 1.079\,g$)

Figure 5.3 captures the critical transition curve when the unbalance mass was engaged. The real-time vibration graph displays the steep upward surge from $0.013\,g$ through the warning zone to $0.758\,g$, peaking at $1.079\,g$. The dashboard immediately updated its status banner to `CRITICAL ALARM` in vibrant red, while the persistence meter saturated at 100%, mirroring the physical Fast Red Strobe on the hardware testbed.

5.6 Visual RGB LED Indicator Truth Table

VibeGuard provides immediate, deterministic physical alerting through an onboard common-cathode RGB LED. Table 5.3 documents the complete truth table of operational modes, color mappings, and flash behaviors.

Table 5.3: VibeGuard physical RGB LED indicator status truth table

Color	Actuation Pattern	System State	Physical Meaning / Action
Solid Green	Continuous ON	`STATE_NORMAL`	Healthy baseline; vibration
Solid Yellow	Continuous ON	`STATE_WARNING`	Stage 1 moderate unbalance
Fast Red Strobe	6.25 Hz (80 ms ON/OFF)	`STATE_ALARM`	Stage 2/3 critical unbalance
Solid Blue	Continuous ON	`STATE_CALIBRATING`	8.0 s Median+MAD baseline
Purple / Magenta	Continuous ON	`SENSOR_FAULT`	SPI communication failure

5.7 Frequency-Domain Harmonic Peak Detection

To confirm that the vibration surge originated from mechanical rotor unbalance rather than random electrical noise, the on-demand Cooley-Tukey FFT was triggered. Table 5.4 presents the extracted spectral peak data.

Table 5.4: FFT spectral peak extraction results across operating states

Operating State	Dominant Bin (k)	Frequency (f_{peak})	Magnitude	Physical Interpretation
Normal Baseline	Bin $k = 48$	150.0 Hz	0.018 g	Low-energy gear mesh tone
Normal Baseline	Bin $k = 3$	9.375 Hz	0.014 g	Residual 1X rotor balance
Unbalance Fault	**Bin $k = 3$ / 4**	**9.38 – 12.5 Hz**	0.224 g	**Synchronous 1X Unbalance**
Unbalance Fault	Bin $k = 6$ / 7	18.75 – 21.88 Hz	0.045 g	Secondary 2X harmonic

Under the unbalance condition, the discrete magnitude spectrum exhibited a sharp, prominent peak centered at the 10.0 Hz fundamental rotational frequency (Bin $k = 3/4$). The discrete magnitude at 10 Hz reached 0.224 g, representing a **16-fold energy increase** over the healthy baseline state at that frequency. A minor secondary peak emerged at the $2X$ harmonic (18.75–21.88 Hz) due to baseplate structural coupling, decisively confirming the physical presence of mechanical unbalance.

5.8 Persistence Filter Response and Shock Rejection

To evaluate system resilience against false alarms, two distinct disturbance scenarios were experimentally evaluated, as documented in Table 5.5.

Table 5.5: Experimental verification of persistence filter and alert response

Test Scenario	Peak a_{vec}	Violations (V)	State Output	Observed Behavior
Transient Shock (Screwdriver Tap)	$0.285\,g$	$1/5 < K$	`STATE_NORMAL`	Jolt rejected; LED G
Sustained Unbalance (Calibrated Mass)	$1.079\,g$	$4/5 \geq K$	`STATE_ALARM`	Prompt alarm; Red S

5.9 Computational and Memory Execution Budget

The entire VibeGuard firmware pipeline was compiled and profiled on the ESP32 hardware:

- **Program Flash Memory:** 314,476 bytes utilized (23.9% of available 1.31 MB application partition).
- **Dynamic SRAM Usage:** 28,492 bytes utilized (8.6% of available 327 KB SRAM).
- **Real-Time Execution Time:** The complete DSP pipeline (Butterworth biquad filtering, autocorrelation speed tracking, $J(t)$ extraction, Huber limiting, and persistence update) executes in under 1.42 **ms** per 160 ms window, consuming $< 1\%$ of single-core CPU capacity.

5.10 Performance Benchmarking with Other Approaches

Table 5.6 compares VibeGuard with alternative industrial and academic vibration monitoring paradigms.

Table 5.6: Comprehensive benchmarking of VibeGuard against alternative approaches

Design Attribute	**Manual Tactile / Stethoscope**	**Cloud Streaming IIoT**	**Continuous Embedded FFT**	**VibeGuard System (Proposed)**
Hardware Complexity	Very Low (Hand tool)	High ($>$ \$500)	Medium ($\approx$ \$100)	**Low (Single MCU + Sensor)**
Detection Latency	Days to Weeks	5 – 30 seconds	1 – 2 seconds	**1.28 seconds**
Network Dependency	None	Continuous Wi-Fi / Cellular	Periodic Wi-Fi	**Zero (Autonomous Edge)**
Power Consumption	0 W	3 – 8 W	2 – 3 W	$<$ **1.5 W**
False-Alarm Rejection	Poor (Subjective)	Moderate	Poor (Threshold flicker)	**High (K-of-M Persistence)**
Substrate Invariance	N/A	Low (Requires model retraining)	Low (Manual recalibration)	**High (Homogeneity of Degree Zero)**
Explainability	Subjective intuition	Black-box cloud	Complex spectra	**Direct Vector RMS + $J(t)$ + FFT**

5.11 Summary

Experimental results confirmed that VibeGuard achieves reliable, low-latency (1.28 s) vibration anomaly detection. The baseline Vector RMS of 0.013–0.060 g jumped by 423% to 1.079 g during severe unbalance, producing a distinct 16-fold spectral energy concentration at the 10.0 Hz fundamental frequency. The deep research substrate-invariant metric $J(t)$ and crest-factor shock limiter completely eliminated Stage 1 yellow red-blinking across a 100-window stability audit while ensuring robust assertion of Fast Red Strobe alarms under dynamic eccentric unbalance.

CHAPTER 6
CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This micro project presented the design, implementation, and empirical validation of **VibeGuard**, a low-cost, edge-resident vibration condition monitoring system engineered for rotating machinery. By integrating an Analog Devices ADXL345 3-axis digital MEMS accelerometer with a 32-bit dual-core 7Semi ESP32-DEVKIT-E microcontroller via a deterministic 5 MHz 4-wire SPI bus, the system successfully executes real-time data acquisition at an 800 Hz Output Data Rate (ODR).

The edge firmware pipeline resolves real-world vibration sensing challenges directly on the embedded microcontroller without relying on remote cloud connectivity:

- Gravitational bias and static mounting tilt are eliminated through in-place zero-mean DC cancellation across rolling 256-sample (320 ms) analysis windows.
- Omnidirectional vibration magnitude is extracted using a true triaxial Vector Root-Mean-Square formulation (V_{RMS}), supplemented by an adaptive autotune threshold ladder (Median + $k \times$ MAD).
- Benign transient mechanical jolts and quantization noise are successfully filtered out using a K-of-M persistence filter ($K = 4, M = 5$), achieving 100% false-alarm rejection during impulse shock testing.
- Physical operational states are reliably differentiated across three distinct regimes: healthy baseline (Stage 0: Solid Green LED, $V_{\mathrm{RMS}} \approx 0.013$–$0.060\,g$), moderate unbalance (Stage 1: Solid Yellow LED, $V_{\mathrm{RMS}} \approx 0.82$–$0.86\,g$), and severe unbalance (Stage 2/3: Fast Red

Strobe at 6.25 Hz, $V_{\text{RMS}} \rightarrow 1.079\,g$).

- An alert latency of only 1.28 seconds was empirically verified, triggering immediate local visual indication and structured telemetry streaming to an interactive web dashboard.
- When triggered, an embedded 256-point Cooley-Tukey Radix-2 FFT with Hann windowing extracts dominant spectral peaks across the 0–400 Hz Nyquist band, confirming a 16-fold energy concentration at the 10.0 Hz synchronous fundamental unbalance frequency.

Operating at an ultra-low power consumption of under 1.5 W, VibeGuard demonstrates that explainable, deterministic machine health protection can be achieved locally on a microcontroller platform, eliminating the high costs, bandwidth demands, and security risks of cloud-streaming architectures.

6.2 Future Scope

While the current semester baseline fulfills all functional and academic deliverables, several future enhancements can be pursued:

1. **Multi-Sensor Spatial Fusion:** Future iterations can incorporate secondary accelerometers at driven-end and non-driven-end bearing locations to study structural vibration attenuation and spatial fault localization.
2. **Advanced Edge Spectral Band Features:** The on-demand FFT module can be expanded to automatically track harmonic fault frequency bands associated with bearing inner/outer race pass frequencies (BPFI/BPFO) and gear mesh harmonics.
3. **Onboard Non-Volatile Event Logging:** Integrating an SPI NOR Flash or MicroSD card interface will allow circular logging of raw vibration waveforms during alarm events for forensic engineering analysis.

4. **Industrial Enclosure and Standardized Field Mounts:** Packaging the electronics into an IP65-rated sealed enclosure with magnetic transducer mounts to evaluate performance across industrial machining and HVAC equipment.

5. **BLE / LoRaWAN Edge Telemetry:** Implementing low-power wireless telemetry protocols to transmit periodic summary metrics to plant supervisory control systems.

CHAPTER 7
CODE

7.1 ESP32 Main Controller Code

The following C++ code snippet represents the core firmware execution loop implemented in the production VibeGuard firmware (`VibeGuard_Autotune_Synchronized.ino`). It implements real-time 800 Hz sensor acquisition over SPI, in-place zero-mean DC cancellation, 3-axis Vector RMS extraction, adaptive threshold evaluation, and K-of-M persistence filtering.

```
void loop() {
    // 1. Acquire 256 samples @ 800 Hz (320 ms window)
    AccelGData g_sample;
    for (size_t i = 0; i < 256; ++i) {
        accel.readG(g_sample);
        x_buf[i] = g_sample.x;
        y_buf[i] = g_sample.y;
        z_buf[i] = g_sample.z;
        delayMicroseconds(1250); // Deterministic 800 Hz pacing
    }

    // 2. Remove DC gravity offset and static mounting tilt
    remove_dc(x_buf, 256);
    remove_dc(y_buf, 256);
    remove_dc(z_buf, 256);

    // 3. Compute Triaxial Vector RMS magnitude
    float v_rms = compute_vector_rms(x_buf, y_buf, z_buf, 256);

    // 4. Update Warning and Alarm persistence queues (K=4, M=5)
    bool is_warn = (v_rms >= g_profile.warning_threshold_g);
    bool is_alarm = (v_rms >= g_profile.alarm_threshold_g);

    update_persistence(is_warn, is_alarm);
    evaluate_condition_state();

    // 5. Update physical RGB LED and transmit telemetry
    applyIndicator();
    transmit_json_telemetry(v_rms);
}
```

Listing 7.1: VibeGuard Core Real-Time Acquisition and DSP Firmware Loop

7.2 Multi-Stage Physical RGB LED Indication Logic

Listing 7.2 details the multi-color RGB LED actuation routine responsible for visually rendering the physical operating states (Normal, Warning, Alarm, Calibration, Fault).

```
void applyIndicator() {
    // 1. Hardware or Sensor Acquisition Fault -> Purple / Magenta
    if (!g_sensor_ready || g_committed_state == ConditionState::
        SENSOR_FAULT) {
        setRGB(true, false, true); // Red + Blue = Purple
        return;
    }

    // 2. Active Baseline Calibration (8.0 s Median + MAD) -> Solid
        Blue
    if (g_vibeEngine.getCalibrationState() != CAL_IDLE) {
        setRGB(false, false, true); // Solid Blue
        return;
    }

    // 3. Physical Operating States
    switch (g_committed_state) {
        case ConditionState::ALARM: {
            // Stage 2/3 Severe Unbalance -> Fast Red Strobe @ 6.25 Hz
            bool strobe = (millis() % 160) < 80;
            setRGB(strobe, false, false);
            break;
        }
        case ConditionState::WARNING:
            // Stage 1 Moderate Unbalance -> Solid Yellow (Red + Green)
            setRGB(true, true, false);
            break;
        case ConditionState::NORMAL:
            // Stage 0 Healthy Baseline -> Solid Green
            setRGB(false, true, false);
            break;
        default:
            setRGB(false, false, true);
            break;
    }
}
```

Listing 7.2: Multi-Stage Physical RGB LED Indicator Actuation

7.3 Substrate-Invariant Rotordynamic DSP Engine Implementation

Listing 7.3 illustrates the deep research rotordynamic DSP pipeline implemented in `VibeGuardEng` showcasing the biquad low-pass cascade, autocorrelation shaft speed tracker, and the dimen-

sionless condition index $J(t)$.

```
// Biquad Direct Form II 4th-Order Butterworth (25 Hz cutoff @ 800 Hz ODR)
float processBiquad(float in, BiquadState& s, const BiquadCoeffs& c) {
    float w = in - c.a1 * s.w1 - c.a2 * s.w2;
    float out = c.b0 * w + c.b1 * s.w1 + c.b2 * s.w2;
    s.w2 = s.w1;
    s.w1 = w;
    return out;
}

// Bounded Autocorrelation Shaft Speed Tracker (7.5 - 14.0 Hz)
float estimateShaftSpeedHz(const float* x, int N) {
    float best_r = -1.0f;
    int best_lag = 80; // 10.0 Hz nominal
    for (int k = 55; k <= 105; ++k) {
        float r = 0.0f;
        for (int i = 0; i < N - k; ++i) r += x[i] * x[i + k];
        if (r > best_r) { best_r = r; best_lag = k; }
    }
    return 800.0f / static_cast<float>(best_lag);
}

// Dimensionless Condition Metric J(t) with Speed-Sag Compensation
float computeMetricJ(float Exy_LF, float Rx, float Ry, float CF, float f1) {
    float c_sag = sq(cal_.f1_0 / f1);
    float term_energy = (Exy_LF / cal_.Exy_LF_0) - 1.0f;
    float term_product = ((Rx / cal_.Rx_0) * (Ry / cal_.Ry_0) * (CF / cal_.CF_0)) - 1.0f;
    return c_sag * (0.55f * term_energy + 0.45f * term_product);
}
```

Listing 7.3: Substrate-Invariant Rotordynamic Metric $J(t)$ Extraction

7.4 Code Walkthrough and Implementation Details

The implementation highlighted in Listings 7.1, 7.2, and 7.3 incorporates several embedded software principles:

1. **Deterministic Hardware Pacing:** Sampling is strictly regulated at 1250 μs intervals using the high-resolution hardware microsecond timer (`delayMicroseconds`), guaranteeing an exact 800 Hz Output Data Rate without timing jitter.

2. **In-Place Signal Preprocessing:** The static DC components (Earth's 1.0 g gravity bias and physical sensor tilt) are computed and subtracted directly within the input sample

arrays using `remove_dc()`, avoiding duplicate memory allocations.

3. **Direct Form II Biquad Filtering:** Cascaded 2nd-order stages suppress gear tooth mesh impacts (150–350 Hz) by over 36 dB while maintaining minimal state register overhead (two delay elements per biquad).

4. **Substrate-Invariant Ratio Metric:** Computing $J(t)$ as normalized quotients guarantees mathematical homogeneity of degree zero, ensuring identical threshold sensitivity regardless of whether the rig rests on wood, acrylic, or steel foundations.

5. **Dual-Stage Persistence Filtering:** Independent circular queues track both warning and alarm violations over sliding 5-window depths ($M = 5$), preventing transient shocks from provoking premature alarm states.

6. **Non-Blocking Strobe Generation:** The 6.25 Hz Fast Red Strobe utilizes modulo arithmetic on `millis()` rather than blocking `delay()` calls, preserving real-time telemetry streaming and sensor sampling continuity.

7. **Zero-Heap Allocation in Steady State:** All sample buffers and filter states are statically allocated in internal SRAM, eliminating dynamic memory fragmentation and guaranteeing long-term continuous uptime.

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